Orphans and Archetypes: Linking Unidentified Ritual Fragments to Hittite Textual Traditions through Quantitative Scribal Tradition Mapping

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Abstract

The Hittite ritual corpus contains a large body of manuscript fragments that resist classification within established textual traditions: Laroche's residual category CTH 470 and the newly digitised EBo fragments, 3,033 manuscripts in the TLHdig 0.3 release, with a median of 10 (CTH 470) and 4 (EBo) legible words. Philological assignment needs a parallel passage, which these "orphans" lack. We treat tradition assignment as a content-based classification problem. Word tf-idf and orthographic sign n-gram features from TLHdig define a reference space of ten ritual and festival traditions, each delimited by the study that argued for it and mapped back to Laroche's CTH numbers. A held-out manuscript is assigned to its correct tradition in 84% of cases (majority baseline 30%), and a ten-word window in 65%. Separating the layers of a text shows that traditions are identified by what is written (logograms, Hittite vocabulary, foreign-language words), not by how it is written: spelling adds nothing beyond the words, and orthographic rates carry only a weak signal that follows compositions and their copies. On CTH 701, where Salvini and Wegner's subgroups provide partial ground truth, the method recovers 57% of subgroup labels (majority 21%) and does not predict script date. When whole compositions are held out, 57 of 78 return to their scholarly group: the Hattian, Ḫuwaššanna and Kizzuwatna groups are recovered, the Hurrian festival groups and Palaic are not. With a calibrated abstention margin, 847 of 2,026 orphans are placed; an open-set test shows that a closed reference must be paired with explicit outside classes before such placements can be read as "this tradition". We interpret the results through Quantitative Scribal Tradition Mapping (QTM), which treats orphans as low-coverage witnesses to incompletely attested archetypes.

Keywords

Hittite, cuneiform, stylometry, fragment attribution, open-set recognition, textual traditions, digital philology

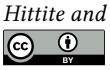
1. Introduction

Emmanuel Laroche's *Catalogue des textes hittites* [1] gave every Hittite composition a number, and with it a place: rituals in one block, festivals in another, texts in Hattian, Luwian and Hurrian in blocks of their own. What could not be placed went into residual categories. CTH 470, "fragments of rituals", is the largest of them, and it has grown with every volume of hand copies since 1971. The Mainz Hethitologie-Portal now keeps Laroche's numbers as a shared address system while moving fragments after joins and adding new texts [2, 3]. The digitisation of the EBo fragments in TLHdig 0.3 [4] adds 1,151 more manuscripts with a median of 4 legible words, most of them filed under CTH 832 (562) and CTH 670 (382), two further residual categories.

The philological route to assigning such a fragment runs through a parallel: a duplicate or a close parallel passage in an identified composition. That route is closed for most orphans by definition. This paper asks whether the content of a fragment, its words and the way they are spelled, can point to its tradition instead, and how far such a pointer can be trusted. It builds on an exploratory study of the Hurrian and Luwian festival texts [5], which mapped about 1,000 manuscripts with tf-idf and t-SNE, examined CTH 701 under the "microscope", and announced as its next step the test of which word groups carry the signal. Here the exploration becomes assignment, with explicit baselines, held-out evaluation and a rule for when not to assign.

Our contributions are:

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1. a reference space of ten Hittite ritual and festival traditions, each defined by the study that argued for it and mapped back to Laroche's CTH numbers, with the CTH ranges that no study claims set aside as test sets (Section 3);
2. a validation of nearest-centroid tradition assignment against majority and permutation baselines, as a function of fragment size, and on CTH 701, where partial ground truth exists, and a leave-one-composition-out test of the scholarly groups themselves (Sections 5.1–5.6);
3. a decomposition of the signal into layers and word classes that separates what is written (cultural content) from how it is written (scribal habit), and an exact split of each decision over them, which shows how much rests on common particles such as *nu* (Sections 5.4–5.5);
4. a placement of the CTH 470 and EBo orphans with a calibrated abstention margin, and an open-set test of whether the method notices a fragment that belongs to no reference tradition (Sections 5.7–5.8);
5. Quantitative Scribal Tradition Mapping (QTM), a framework that reads the results in terms of witnesses and archetypes (Section 6).

All code, the label snapshot and the outputs are public, and every number in this paper is generated from the experiment outputs (Section 9).

2. Background and related work

Traditions and the catalogue. Laroche's catalogue sorts by genre first and by language second, while later scholarship defines a ritual tradition by the origin of its practitioners, its cult and its ritual language. One tradition therefore ends up scattered over several CTH ranges. The Kizzuwatna rituals occupy CTH 471–500, but the ritual of the Kizzuwatnean Old Woman Maštigga is filed with the named practitioners at CTH 404 [6]; the rituals of the AZU priest (CTH 701) sit in the Hurrian block rather than with the Kizzuwatna rituals [7]; and the named-practitioner rituals of CTH 390–469 come from Arzawa, the Lower Land and Kizzuwatna alike [8, 9]. The festival texts have been regrouped by cult geography [5, 10, 11] and, in the Hethitische Festrituale (HFR) project, into edition modules such as the Luwian and the Palaic festivals [12, 13]. Surveys of Hittite religion frame all of these [14, 15].

Scribal habit and its measurement. Hittite scribes copied, excerpted and re-edited ritual texts over two centuries, and their writing habits (logographic writing, plene spelling, sign choice) vary with date, school and individual [16, 17, 18, 19, 20]. Whether a habit marks a tradition, a scribal circle or a date is therefore an empirical question, and one this paper tests rather than assumes. Stylometry offers the tools: frequency-based representations, distance measures and attribution protocols with held-out evaluation [21, 22, 23]. Two of its known problems are central here. Short samples degrade attribution sharply [24], and a closed-set classifier always names one of its candidates, even when the true author is not among them [25]; the latter is the problem of open-set recognition [26].

Computational work on cuneiform. Machine learning on cuneiform has addressed transliteration and translation [27, 28], language and dialect identification [29], the matching of fragments to known compositions by sign sequences [30], and palaeographic style [31]. Fragment matching finds a known text; it does not place a fragment whose parallels are lost. All of these studies are closed-set. To our knowledge, no published study uses content features to assign Hittite fragments to religious traditions, or asks when a fragment fits none of the candidates.

3. Data

Corpus. We use the TLHdig Beta 0.3 release [4], converted to one row per word following [28]. Each word carries its normalised form, its transliteration split into signs, its language layer and, where TLHdig provides a morphological analysis, a word class. The release is downloaded by script and checked by its

MD5 sum, so the token table can be rebuilt from the public data alone. As in [5], reference manuscripts must have at least 17 legible words.

Orphans. Orphans are all CTH 470 manuscripts and all EBo manuscripts, wherever TLHdig files them (1,919 and 1,151 manuscripts; 3,033 in total). The experiments load the reference CTHs, CTH 470 and the two pools CTH 670 and CTH 832; these contain 2,899 orphans, of which 2,026 have at least five legible words and are placed (median 7 words over all loaded orphans). Orphans are never part of the reference.

Reference traditions. The abstract that announced this study proposed CTH 670, 701 and 832 as the reference. In TLHdig, however, CTH 670 and 832 are where the EBo fragments are filed, so they are pools of orphans rather than traditions; as a three-class problem they are also dominated by CTH 670 (majority baseline 78%). We therefore define the reference from the literature (Table 1). The Hurrian festival groups follow Table 3 of [5]. Its single Luwian group is split as the HFR edition modules split it, into the festivals of Ištanuwa and Lallupiya and those of Ḫuwaššanna, and HFR's Palaic festivals are added [12]. The ritual traditions are the Kizzuwatna rituals as delimited by Miller [6]; the western (Arzawa) rituals, that is, the augur rituals [9] and the rituals whose colophon names Arzawa or Ḫapalla as the practitioner's home; the Luwian rituals of Puriyanni, Kuwattalla and Šilalluḫi (CTH 758–763), which Yakubovich and Mouton place in the Lower Land rather than Kizzuwatna [32]; and the Hattian cult layer of Klinger [33], whose texts we take from the citations of his study in the Mainz Konkordanz [3]. The Ambazzi rituals (CTH 391, 429, 463) show Luwian influence but no stated link to Arzawa [8], so they are not part of the western group and are placed with the residue; Section 5.2 tests that decision. The published grouping of [5] with two CTH-range controls (CTH 471–500 and CTH 390–469) is run as the baseline.

Two sets leave the reference and are placed like orphans. CTH 500, "fragments of festival and incantation rituals from Kizzuwatna", is a running list of fragments rather than a composition, and HFR counts it among its festival texts; and the texts of CTH 390–469 that no defining study claims form a residue. Both become test sets (Section 5.7).

CTH 701 labels. The subgroup labels of CTH 701 (Salvini and Wegner's groups a–f with their sub-numbered parallels [7]) and the script dates (Middle and New Script) come from the Mainz Konkordanz [3], parsed by script and frozen as a dated snapshot. Checked against the working sheet of [5], the groups agree for 120 of 121 shared fragments; the one difference is a missing group letter in the Konkordanz (KBo 64.168, listed as "701.$_2$.I"), corrected to b$_2$ in a documented table.

4. Method

Figure 1 summarises the procedure and the tests reported in Section 5. Figure 2 follows one orphan, KBo 41.22, through the procedure, with the actual values at each step.

Features. Two representations are built for every manuscript [38]. The word space holds tf-idf weights of normalised word forms, and represents what is written. The sign space holds tf-idf weights of sign n-grams (n = 1–3) taken within words, and represents how it is written. The combined space concatenates both. Weights are sublinear, rows are L2-normalised, and features occurring in fewer than two manuscripts are dropped.

Assignment. Each tradition is represented by the normalised centroid of its manuscripts, and a fragment is scored by its cosine similarity to every centroid. It is assigned to the most similar tradition when its lead over the runner-up (the *margin*) exceeds a threshold, and left unassigned otherwise.

Table 1
Reference traditions, the studies that define them, and the number of reference manuscripts with 17 or more legible words. CTH numbers are those of the Portal CTH [2].

Tradition	CTH	Defining studies	Criterion	*n*
Kizzuwatna–Lower Land (Hurrian)	628, 699–700, 711–713, 719, 722	Gordin et al. [5] Table 3; Wegner and Salvini [34]	Hurrian festivals whose cult geography is Kizzuwatna and the Lower Land, among them the *(ḫ)išuwa*-festival (CTH 628)	318
Syro-Mesopotamia (Hurrian)	698, 701, 714–715, 721	Gordin et al. [5] Table 3; Salvini and Wegner [7]	rituals of the AZU priest (CTH 701) and the Ištar/Šawuška cults of Nineveh (714–715) and the Amanus (721)	167
Mixed (Hurrian, Syrian, Luwian)	704–706, 720	Gordin et al. [5] Table 3	festivals with mixed Hurrian/Syrian/Luwian cult elements	200
Luwian festivals (Ištanuwa, Lallupiya)	665, 771–773	HFR-Team [12]; Görke [13]; Hutter [35]	HFR edition module "Luwian festivals": cult towns Ištanuwa and Lallupiya	52
Ḫuwaššanna festivals	690–694	HFR-Team [12]; Hutter [35]	cult of Ḫuwaššanna of Ḫubešna in the Lower Land (Laroche's titles); kept apart from Ištanuwa by HFR	189
Palaic festivals	750–754	HFR-Team [12]; Görke [13]	HFR edition module "Palaic festivals"	34
Kizzuwatna rituals	404, 471–499	Miller [6]; Strauß [36]; Görke [37]	practitioners from Kizzuwatna, Hurro-Luwian ritual language; Laroche's block plus Maštigga (CTH 404); the fragment collection CTH 500 excluded	165
Arzawa / western rituals	394, 402, 406, 410, 425, 433	Bawanypeck [9]; colophons	augur rituals (425, 433) and the rituals whose colophon names Arzawa or Ḫapalla as the practitioner's home (394, 402, 406, 410); the Ambazzi rituals (Christiansen 8) left the group, as they show Luwian influence but no stated link to Arzawa	11
Luwian rituals (Puriyanni, Kuwattalla)	758–763	Yakubovich and Mouton [32]	Luwian rituals of Puriyanni, Kuwattalla and Šilalluḫi, placed in the Lower Land, not Kizzuwatna	10
Hattian cult layer	591, 627, 725–729, 731–746, 820–821	Klinger [33]	texts cited by StBoT 37 in the Mainz Konkordanz: Monatsfest, KI.LAM, Hattian and Hattian–Hittite texts	480

Evaluation. Validation is leave-one-manuscript-out: each reference manuscript is removed from its own centroid, which can be done in closed form, and reassigned. We report accuracy and per-tradition recall against two baselines: always guessing the largest tradition (majority), and the same procedure with shuffled labels (200 permutations, which also give a p-value). The idf weights are fitted on the whole reference set (transductive idf).

Fragment size and abstention. To learn how small a fragment can be, manuscripts with at least 40 words (at most 40 per tradition) are held out in turn and cut into ten random windows of k words, $k \in \{3, 5, 8, 10, 17, 25, 40\}$. The abstention threshold is the smallest margin at which the held-out windows reach 80% precision; a bootstrap over the windows gives its 90% interval.

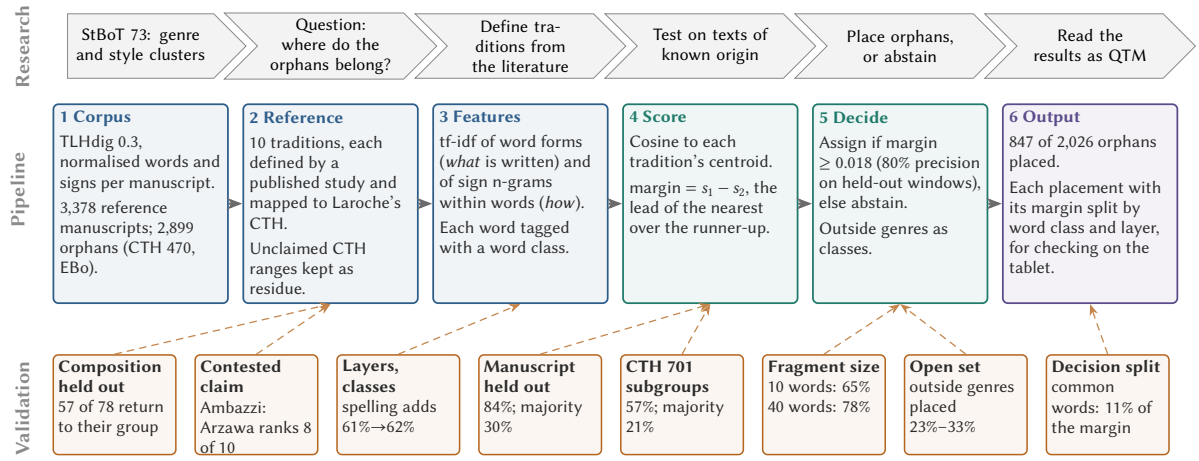


Figure 1: Research process (top), pipeline (middle) and the tests that check each step (bottom; dashed arrows point to the step a test checks). Numbers are those of the main run and are generated from the experiment outputs.

Layers and word classes. To separate cultural content from scribal habit, the word space is split into layers and each layer is evaluated alone: logograms (Sumerograms and Akkadograms), Hittite vocabulary, foreign-language words (Hurrian, Luwian, Hattian, Palaic), and the 392 words attested in more than one spelling, once with their spelling kept and once collapsed to the normalised form. If the spelling carried tradition beyond the word, the first would beat the second. Seven per-manuscript orthographic rates (shares of logograms, Akkadograms, phonetic complements, determinatives, plene spellings, CVC signs and *ú*) are tested with a 10-fold linear discriminant analysis. Following the categories of Table 4 in [5] (agents, deities, function words, foreign words, realia and verbs), with realia narrowed to physical things and the other nominal words grouped by meaning (places; animals, body parts and nature; abstract nouns; adjectives; measures), assigned as described below, we evaluate each word class alone and the text without it. Distinctive features per tradition are ranked by weighted log-odds with an informative Dirichlet prior [39], on manuscript frequencies.

How a word gets its class. No word was classed by hand. Each class is derived by fixed rules (`qtm/wordclass.py`) from what TLHdig's editors recorded for the word: its determinatives, its language, and its morphological analyses, of which the editor may have selected one. The rules apply in order, and the first that fits decides (Table 2). A determinative decides first, because deities and persons cut across parts of speech: DINGIR or ^{D} makes a deity, ^{LÚ}, ^{MUNUS} or a personal-name marker an agent, and so do a short list of agent logograms (LUGAL, SANGA, …); a compound logogram headed by an object sign is an object (GIŠ.^{D}INANNA, 'lyre'). A syllabically written Akkadian preposition (*ina*, *ana*, *ištu*, *itti*, *aššum*) is a function word. A word in Hurrian, Hattian, Luwian or Palaic is foreign; a logogram without a language tag of its own (KI.MIN in a Hurrian line) is not, and goes on to its analysis. Otherwise the editor's selected analysis gives the part of speech: a person-marked form is a verb; conjunctions, particles, pronouns, adverbs, preverbs and postpositions are function words; any other case-marked form is nominal. An Akkadographic preposition that TLHdig stores in one word with its noun (A-NA ^{GIŠ}BANŠUR) is split, and each part classed on its own. When no analysis is selected, the word takes the class on which all its candidate analyses agree, or else the class the editors selected for the same form elsewhere, if at least five selections agree in at least 90% of cases. A word with no analysis at all takes its class from a determinative (^{URU} a place, ^{MUŠEN} an animal, ^{GIŠ}, ^{NINDA} and the like an object); a broken word that ends in an enclitic chain (*-ša-an*) is a function word, and one that ends in *-anzi* a verb.

A nominal word is then classed by meaning. Names take their class from TLHdig's stem class (gods, persons, places). Every other noun and adjective takes the class of its German gloss in TLHdig, looked up in a list that assigns each of the 5,494 glosses one meaning class (`labels/gloss_lexicon.tsv`):



Figure 2: A worked example: the orphan KBo 41.22 from its text to its placement. (1) The legible words, tinted by word category (dashed: a low-confidence or unclear class). (2) Each word becomes a word-form feature and sign-sequence features, weighted by tf-idf. (3) Cosine similarity to the average vector of each tradition. (4) The lead of the nearest tradition over the runner-up, against the calibrated margin. (5) The lead split over the features that make it. (6) The same split on the text, and the closest single manuscripts as leads for the philologist. All values are generated from the experiment outputs.

realia for artifacts, buildings, ritual materials, food and plants, and otherwise animals, body parts, nature, abstract nouns, adjectives or measures. This list is the one step not derived from the editors' records: it was drafted with a language model (Claude), is committed as a fixed file that can be corrected by hand, and has not yet been checked. When the glosses of a word's candidate analyses fall in different classes, it is realia if one of them is, and ambiguous otherwise.

Each label records the rule that set it, and so a confidence: high for a determinative, a language tag, an Akkadian preposition, a selected analysis or a split preposition; medium for the agent list, unanimous candidates, the editors' choice elsewhere or a determinative without analysis; low when candidates disagree, for the rules for broken words, and when no analysis exists. Of the 320,638 legible words of the texts we use, 63% are classed with high confidence, 25% with medium and 12% with low; 9% have no analysis (mostly broken word ends) and 3% stay ambiguous. Participles count as verbs. A bare 10 or 30 stays a number, as the editors select it in most cases; the Storm-god and the Moon-god, written $^{\text{D}}10$ and

Table 2
How a word gets its class: one example per rule, with TLHdig's analysis as the editors recorded it (glosses are TLHdig's, in German) and the class and confidence it receives. The rules apply in this order.

Rule	Word	TLHdig analysis	Class	Conf.
a deity determinative or DINGIR	*DINGIR^{MEŠ}*	2 candidates: 'Göttlichkeit' NOM.SG(UNM)…; 'begei…	deity	high
a person determinative (LÚ, MUNUS, m, f)	*^{LÚ}SANGA*	none	agent	high
a listed agent logogram	*LUGAL*	4 candidates: 'Šarrumma' …:D/L.SG…; 'Šarrumma' …:…	agent	medium
a syllabic Akkadian preposition	*a-na*	language tag AKK	function	high
the word or line is not Hittite	*pa-ap-pé-en-na*	selected: 'Berg' HURR.ABS.PL…	foreign	high
the editor selected one analysis	*e-ku-uz-zi*	selected: 'trinken' 3SG.PRS	verb	high
the editor selected one analysis	*nu*	selected: CONNn	function	high
preposition split from its noun	*IŠ-TU*	noun: 'Schale' …:ABL…	function	high
all candidate analyses agree	*na-aš-ta*	4 candidates: CONNn=OBPst; CONNn=PPRO.3SG.C.ACC=O…	function	medium
editors' choice for the same form elsewhere	*ar-ḫa*	6 candidates: 'stehen' 1SG.PRS.MP; 'weg' ADV; …	function	medium
a noun: its meaning class from its gloss	*ḪA-AT-TI*	selected: 'Ḫattuša' GN.NOM.SG(UNM)…	place	high
an object determinative or compound head	*NINDA.SIG^{MEŠ}*	1 candidate: 'Flachbrot' NOM.SG(UNM)…	realia	medium
candidate analyses disagree	*GAL*	2 candidates: 'Großer' NOM.SG(UNM)…; 'Becher' NOM…	ambiguous	low
broken word ending in an enclitic chain	*-ša-an*	none	function	low
broken word ending in 3rd person plural	*-an-zi*	none	verb	low
no analysis (broken word)	*-zi*	none	unknown	low

^{D}30, are deities by their determinative.

The rules were checked twice. An audit of the rules against the examples they produced corrected ten systematic errors (among them A-NA with its noun read as one function word, possessive suffixes taken for the head, GAL counted as an agent, and logograms in foreign-language lines counted as foreign); 83,814 and 4,310 words changed class. The first author then checked by hand a sample of 178 words, drawn evenly from the three confidence levels. The rules were right for 93% of high-confidence words (95% interval 84%–97%), 88% of medium (78%–94%) and 70% of low (57%–80%); weighted by the share of each level in the corpus, 89% of words are classed correctly. The errors led to 5 further rules (the rules for broken words and for object determinatives among them), which changed 2% of words. On the same sample the revised rules reach 94%; as they were written with this sample in view, that figure is an upper bound. The audit scored all nominal words as one class, so it does not cover the split by meaning, which together with the fixes changed the class of 19% of words.

Open set. The margin rule asks only which tradition is nearest. To measure how often it is fooled, we score texts whose genre is known and lies outside the reference, taken from Laroche's own genre blocks: myths, prayers and hymns, oracles, historical texts and treaties, instructions, cult inventories [40], festivals of traditions not in the reference, and the Luwian rituals of Taurisa (CTH 764–766), a further Luwian ritual tradition outside the reference. Reference and outside texts are cut into windows and scored in five folds. We measure the share of outside windows that pass the margin, how well the top similarity separates reference from outside windows (AUROC), and a *genre gate* in which the

Table 3
Leave-one-manuscript-out accuracy of tradition assignment for three reference setups. Majority: always the largest class; shuffled: mean of 200 label permutations.

Reference setup	classes	n	majority	shuffled	words	signs	both
CTH 670 / 701 / 832 (abstract)	3	1,302	78%	53%	88%	89%	**88%**
StBoT 73 groups + CTH-range controls	6	1,676	23%	18%	70%	72%	**72%**
Literature-defined traditions (this paper)	10	1,626	30%	17%	82%	84%	**84%**

Table 4
Per tradition: reference manuscripts, leave-one-out recall in each space, orphans assigned (Section 5.7), and QTM coverage of the 200 core terms by those orphans (Section 6).

Tradition	n	words	signs	both	orphans	coverage
Kizzuwatna–Lower Land (Hurrian)	318	81%	82%	82%	68	94%
Syro-Mesopotamia (Hurrian)	167	71%	67%	68%	60	92%
Mixed (Hurrian, Syrian, Luwian)	200	64%	68%	68%	21	68%
Luwian festivals (Ištanuwa, Lallupiya)	52	81%	83%	83%	8	54%
Ḫuwaššanna festivals	189	88%	90%	93%	12	58%
Palaic festivals	34	82%	82%	82%	3	28%
Kizzuwatna rituals	165	87%	92%	92%	513	100%
Arzawa / western rituals	11	73%	55%	64%	11	52%
Luwian rituals (Puriyanni, Kuwattalla)	10	50%	40%	40%	21	62%
Hattian cult layer	480	90%	94%	95%	130	98%

outside genres become classes of their own.

QTM coverage. For each tradition, the *archetype* is approximated by the 200 highest-weighted terms of its centroid in the combined space, and its *coverage* is the share of those terms attested in at least one orphan assigned to it (Section 6).

5. Results

5.1. Validation: do the traditions hold together?

Table 3 compares the three reference setups. The abstract's CTH 670/701/832 setup reaches 88%, but its majority baseline is already 78%, so it shows little. The published groups with CTH-range controls reach 72% over six classes. The literature-defined traditions reach 84% over 10 classes, against 30% for the majority baseline and 17% for shuffled labels ($p = 0.005$). Words and signs perform alike (82% and 84%).

Per-tradition recall (Table 4) ranges from 40% to 95%. The HFR split holds: the Ḫuwaššanna festivals (93%) and those of Ištanuwa and Lallupiya (83%) are kept apart by the texts themselves, so the single Luwian group of [5] joined two cult traditions. The weakest tradition is also the smallest: Luwian rituals (Puriyanni, Kuwattalla) has 10 manuscripts long enough to test. Removing all logograms lowers accuracy only from 82% to 76% (words) and from 84% to 81% (signs), so the signal does not rest on the logographic cult vocabulary alone.

5.2. Does the method recover the scholarly groups?

Leave-one-manuscript-out accuracy is flattered by duplicates: copies and fragments of the same composition stay in the reference when one of them is held out. A stricter test removes a whole composition (all manuscripts of one CTH number), rebuilds the centroids, and asks whether the pooled composition returns to the group that scholarship assigned it to. 57 of 78 compositions do, and 949 of 1,627

Table 5
Leave-one-composition-out recovery of the scholarly groups: compositions (CTH numbers) and manuscripts assigned to their own group when the whole composition is held out.

Tradition	compositions	manuscripts
Kizzuwatna–Lower Land (Hurrian)	4 of 8	97 of 318
Syro-Mesopotamia (Hurrian)	0 of 5	3 of 167
Mixed (Hurrian, Syrian, Luwian)	3 of 4	113 of 201
Luwian festivals (Ištanuwa, Lallupiya)	2 of 4	14 of 52
Ḫuwaššanna festivals	5 of 5	170 of 189
Palaic festivals	1 of 5	4 of 34
Kizzuwatna rituals	16 of 18	115 of 165
Arzawa / western rituals	1 of 2	1 of 11
Luwian rituals (Puriyanni, Kuwattalla)	0 of 2	0 of 10
Hattian cult layer	25 of 25	432 of 480

manuscripts (58%) are assigned correctly when their composition is unseen (Table 5). The Hattian cult layer (25 of 25), the Ḫuwaššanna festivals (5 of 5) and the Kizzuwatna rituals (16 of 18) are recovered as groups. The Hurrian festival groups (7 of 17 together), the Palaic festivals (1 of 5) and the small western (1 of 2) and Luwian ritual groups (0 of 2) are not: their manuscripts resemble each other because they share compositions, not because the compositions share a profile that sets them apart. With CTH 701 held out, its manuscripts go to the Kizzuwatna rituals (0 of 124 to its own group). We read these groups as claims the present features do not support, not as claims the evidence refutes.

A contested attribution. The Ambazzi rituals (CTH 391, 429, 463) have been read as western because of their Luwian elements [8]. We pool their 6 manuscripts and score them against every tradition, comparing each tradition's lead over the best other with the same statistic for random sets of the same size drawn from the 308 unclaimed ritual manuscripts (CTH 390–469). Arzawa ranks 8 of 10 ($p = 0.18$). The closest tradition is the Kizzuwatna rituals, for 5 of the 6 manuscripts, but no closer than random unclaimed rituals ($p = 0.66$). Of the 20 words most distinctive of the western group, 1 occurs in the Ambazzi texts (*Éri*). On this evidence the Ambazzi rituals belong to the generic ritual residue, which is why they are not in the western reference.

5.3. How small can an orphan be?

Accuracy rises steeply with fragment size and then flattens (Figure 3). A three-word window is assigned correctly in 50% of cases, a ten-word window in 65%, and a forty-word window in 78%, against a chance level of 10%. Sign n-grams are slightly better than words on short windows (64% and 62% at ten words). The abstention margin calibrated on these windows is 0.018 (90% interval 0.018–0.019); above it, 80% of windows are correct, and 66% of them pass.

5.4. What identifies a tradition: what is written, or how?

Table 6 evaluates each layer alone. Content carries the identity of a tradition: logograms reach 69%, Hittite vocabulary 72%, and foreign-language words 88% in the 396 manuscripts that have them. The variable words reach 62% with their spelling kept and 61% with it collapsed: the spelling choice adds nothing beyond the words themselves. The seven orthographic rates are a weak but real signal, 40% against 29% for shuffled labels ($p = 0.0099$).

The word classes of [5] point the same way (Table 7). Removing function words, verbs or agents costs almost nothing (82% with all words, 82% without function words, 81% without verbs): these classes form a layer shared across traditions. Alone they still reach 54% and 56%, well above the majority baseline, so the formulae differ between traditions, but less than agents (66%), realia (59%) or foreign words (84%). Removing foreign words costs the most (78%). The previous study suggested that shared Akkadograms

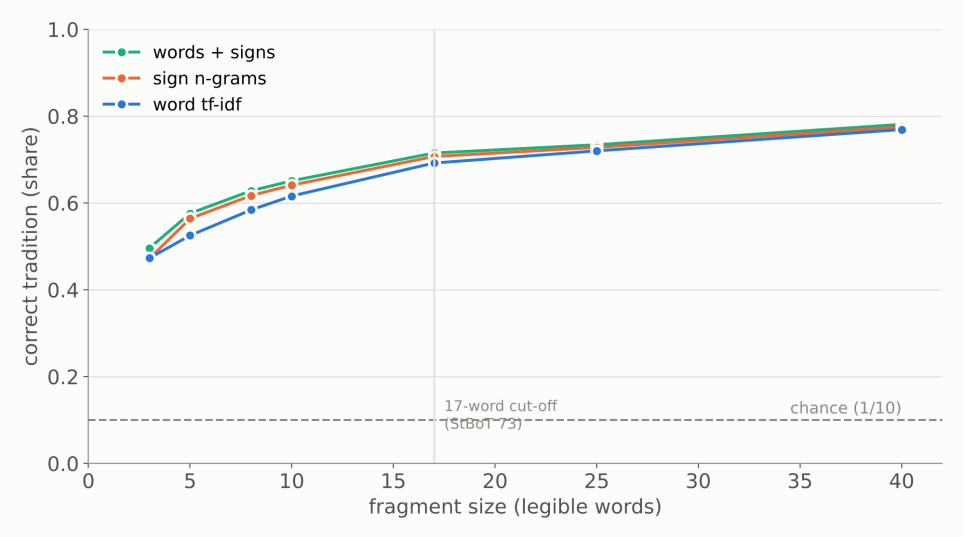


Figure 3: Accuracy of tradition assignment for held-out windows of k legible words. The vertical line marks the 17-word cut-off of [5].

Table 6
Each layer of a text alone: leave-one-manuscript-out accuracy over the reference traditions. n: manuscripts with at least one feature of the layer.

Layer	n	majority	shuffled	accuracy
Logograms (Sumero- and Akkadograms)	1,561	27%	13%	**69%**
Hittite vocabulary	1,556	27%	15%	**72%**
Foreign-language words (Hurrian, Luwian, Hattian, Palaic)	396	50%	24%	**88%**
Variable words, spelling collapsed	1,537	28%	12%	**61%**
Variable words, spelling kept	1,534	28%	12%	**62%**
Seven orthographic rates (LDA, 10-fold)	1,627	30%	29%	**40%**

among function words and verbs might reveal a writing habit [5]; at the scale of traditions, that layer is common ground rather than a marker.

Cultural and scribal profiles. The distinctive features of each tradition (Appendix A) are ritual materials, offerings, performers and deities: AZU, EREN and SÍSKUR in the Syro-Mesopotamian group, SÍG and ZA.GÌN in the Kizzuwatna rituals, LUGAL and DUMU.É.GAL in the Hattian cult layer. Spelling habits behave differently. The plene spelling *me-ma-a-i* appears in 18 of 46 Kizzuwatna–Lower Land manuscripts, all of them CTH 628, 712, and *pé-e-ra-an* in 6 of 21 Ištanuwa and Lallupiya manuscripts, all CTH 665. Spelling habits attach to compositions and their copies, not to whole traditions. Inside CTH 701, the orthographic rates do not predict script date either (79% against a majority of 83%), so there they are not merely a proxy for date. The per-tradition rates are in Appendix B.

5.5. What is a fragment judged on?

Looking at individual placements raises an obvious worry: the most common words of the corpus, the particle *nu* above all, weigh heavily in a fragment's similarity to its tradition. Since the score is a cosine, the decision between the nearest tradition and the runner-up splits exactly over the features, $\cos(x, c_1) - \cos(x, c_2) = \sum_f x_f (c_{1,f} - c_{2,f})$, and every feature occurs in words whose class is known (the categories of [5]). So each decision can be divided into word categories and into layers: the word form, its syllabic signs, and its logograms. Figure 4 compares the two quantities for the 1,372 correctly placed reference manuscripts.

Similarity and decision differ. Features attested in at least a quarter of the reference texts (*nu* and

Table 7
Word classes: accuracy with only that class, and with everything except it, in the word and sign spaces. Majority baseline for the class alone.

			class alone		without class	
Class	*n*	majority	words	signs	words	signs
Agents	1,216	31%	66%	70%	80%	82%
Deities	1,207	23%	57%	57%	80%	82%
Function words	1,539	27%	54%	53%	82%	84%
Verbs	1,479	27%	56%	53%	81%	83%
Realia	1,471	26%	59%	63%	80%	83%
Places	626	32%	59%	53%	81%	84%
Animals	542	24%	32%	34%	82%	84%
Body parts	552	26%	45%	41%	82%	84%
Nature	306	28%	45%	43%	82%	84%
Abstract nouns	421	26%	53%	47%	81%	84%
Adjectives	831	27%	48%	44%	82%	84%
Foreign-language words	621	32%	84%	82%	78%	79%

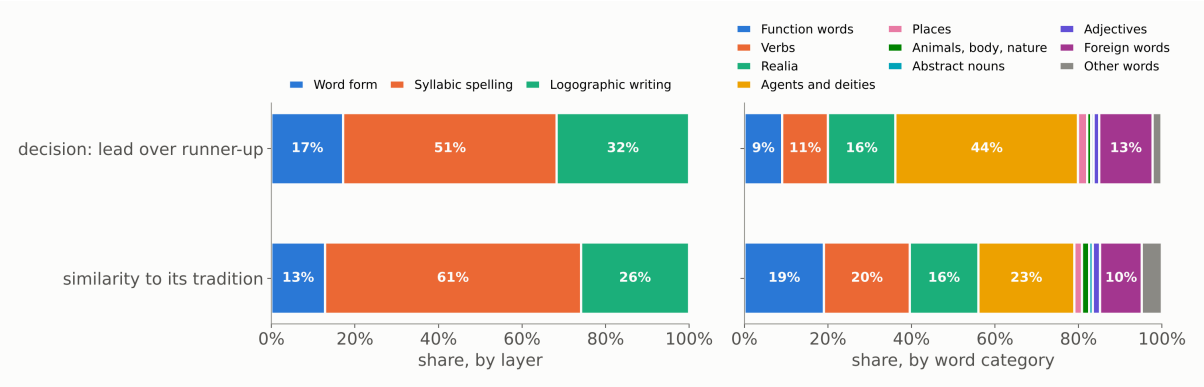


Figure 4: Similarity to the own tradition against the decision (lead over the runner-up), by layer and by word category, for the correctly placed reference manuscripts.

single signs such as *a* and *i*) make up 47% of the similarity but only 11% of the decision, because they are shared by every centroid and largely cancel. The 25 most frequent words carry 15% of the similarity and 2% of the decision. Deleting the 100 most frequent words (44% of all word tokens: A-NA, nu, 1, tai, ŠA, GAL, anta, paršia, našta, 2, ...) leaves leave-one-out accuracy at 83% (84% with them), and deleting 200 at 83%. For whole manuscripts, then, the particles look decisive but are not.

The decision rests on content words: agents and deities 44%, realia 16%, foreign words 13%, verbs 11%, function words 9%, and the other nouns (places, animals, abstract nouns, adjectives) together 5%. Each tradition has its own decisive category (Table 8). The Ḫuwaššanna and Palaic festivals rest on agents and deities (79%, 60%), Palaic also on Palaic words (32%). The Kizzuwatna rituals rest on realia (26%: SÍG, SISKUR, MUNUS-ŠU, MUNUS-ŠU-GI, EN). The Ištanuwa and Lallupiya festivals rest on agents and deities (35%) and on their own place names (28%). Function words decide only where a formula is particular to a group: in the Mixed group (31%) the repetition mark KI.MIN and *EGIR-anda-ma* belong to how its festival lists are laid out. The common features, the negative column of the table, often argue against the small traditions; the Hattian cult layer is the exception, where they carry 25% of the decision and the leading features include single syllabic signs (LUGAL, i, a, LÚ, ú). We take this to be its sign distribution (plene vowels in particular, Appendix B), and so the one tradition where a scribal feature in the narrow sense helps decide. This reading is our inference from the features, not a separate test: the orthographic rates of Section 5.4 are per-text rates, not features of the cosine, and cannot be split into the decision this way.

Table 8
What the decision rests on, per tradition: share of the lead over the runner-up by word category and by layer, and the share carried by common features (in ≥25% of reference texts), over the correctly placed manuscripts (*n*). Shares can be negative when a category argues for the runner-up. "Other" are the other nouns (places, animals, body parts, nature, abstract nouns, adjectives); "other words" (not shown) completes the categories to 100%. Traditions with fewer than ten manuscripts are unstable.

		word category						layer			
Tradition	*n*	funct.	verbs	realia	agents	other	foreign	form	syll.	logo.	common
Kizzuwatna–Lower Land (Hurrian)	261	15%	17%	28%	16%	8%	8%	19%	33%	48%	27%
Syro-Mesopotamia (Hurrian)	113	21%	8%	37%	12%	5%	17%	21%	56%	23%	8%
Mixed (Hurrian, Syrian, Luwian)	135	31%	3%	4%	43%	0%	20%	22%	39%	39%	0%
Luwian festivals (Ištanuwa, Lallupiya)	43	0%	2%	19%	35%	34%	9%	18%	37%	45%	−26%
Ḫuwaššanna festivals	175	2%	8%	12%	79%	0%	0%	17%	51%	32%	4%
Palaic festivals	28	−3%	−6%	16%	60%	1%	32%	17%	66%	17%	−31%
Kizzuwatna rituals	151	14%	10%	26%	12%	23%	4%	22%	50%	28%	2%
Arzawa / western rituals	7	−30%	−1%	102%	51%	0%	0%	37%	−18%	81%	−69%
Luwian rituals (Puriyanni, Kuwattalla)	4	−5%	−13%	1%	−28%	22%	0%	49%	17%	34%	−48%
Hattian cult layer	455	0%	18%	6%	47%	5%	24%	11%	64%	25%	25%
All correctly placed	1,372	9%	11%	16%	44%	5%	13%	17%	51%	32%	11%

Word categories and scribal features. By layer, syllabic signs carry 51% of the decision, logograms 32% and word forms 17%. This does not make spelling the main signal: a syllabic sign n-gram carries the word it writes as well as the way it is written. The spelling choice alone is isolated in Section 5.4, where keeping or collapsing the variant spellings changes accuracy from 61% to 62%. Word categories identify traditions; spelling and logographic habits, apart from the Hattian case, attach to compositions and copies. The distinction the previous study drew between what is written and how it is written [5] holds at the level of the decision as well.

When common words do decide. For short fragments the worry is justified. The share of the decision carried by common features falls with length, from 28% for orphans of 5–10 words to 13% above 50 words, and 227 of the 847 placed orphans rest mostly on them. In windows cut from reference manuscripts, such placements are right 54%, 56% and 62% of the time for 5, 10 and 20 words, against 76%, 81% and 84% for the rest. Removing the words themselves changes few placements: without function words 655 of the 847 placed orphans keep their tradition, 6 move to another and 186 fall below the margin; without the 25 most frequent words the figures are 688, 7 and 152. Common words thus rarely send a fragment to the wrong tradition, but they often supply the margin that lets it be placed at all. We therefore report the decomposition with every placement, and treat one that rests mostly on common features as weak whatever its margin.

5.6. CTH 701 as a test bed

Salvini and Wegner's subgroups of the AZU rituals provide partial ground truth. On 82 manuscripts in 8 subgroups, the method recovers 57% of labels (words 51%, signs 58%) against a majority baseline of 21% ($p = 0.005$; Figure 5). The same procedure does not predict script date: 73% against a majority of 83% ($p = 0.5323$). The subgroups are therefore not script-date clusters. Removing the Hurrian and other foreign words lowers subgroup accuracy from 51% to 47%, the largest drop of any class, yet the remainder is still more than twice the majority baseline: the recitations carry the subgroups, but the Hittite frame text differs between them as well. The 53 unclassified and group-f fragments are placed among groups a–e as candidate attributions in the released tables.

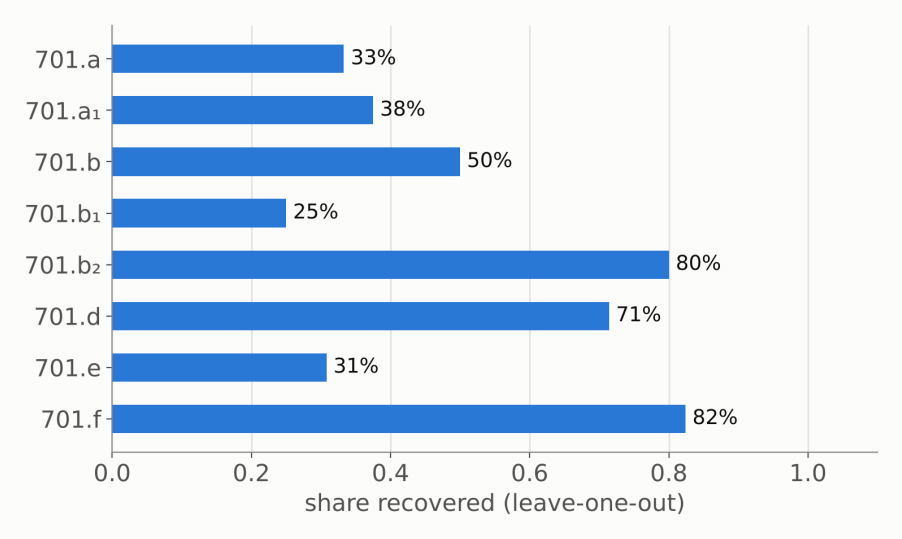


Figure 5: Recall of Salvini and Wegner's CTH 701 subgroups, leave one manuscript out.

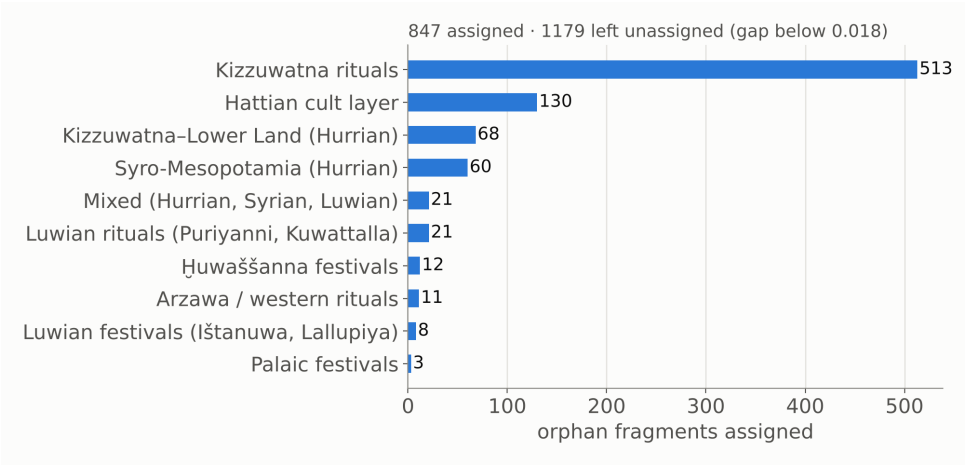


Figure 6: Orphans (CTH 470 and EBo, five or more legible words) assigned to each tradition above the calibrated margin.

5.7. Orphans placed

With the calibrated margin, 847 of the 2,026 orphans are assigned and 1,179 are left unassigned (806–847 assigned across the margin's 90% interval; Figure 6). The Kizzuwatna rituals receive 513 of them (61%).

The two test sets show why that number needs care. The CTH 390–469 residue is placed mostly with the Kizzuwatna rituals (138 of 169 assigned), although no study assigns these texts there: the Kizzuwatna rituals are the only large ritual class, so the placement follows genre before origin. CTH 500 behaves differently. Only 53 of its 132 assigned fragments go to the Kizzuwatna rituals and 78 to the Hurrian festival groups, which agrees with HFR's treatment of CTH 500 as festival material [12].

5.8. Does a fragment fit any known tradition?

Table 9 shows that the margin rule is fooled by texts from outside the reference. At 10 words, it still places 23%–33% of the windows from the six other genres in some tradition, and 60% of the windows from festivals of other traditions. A floor on similarity separates other genres fairly well (AUROC 0.78–0.88) but not other festivals (0.58). A genre gate works better: with the outside genres as classes of their own, 87%–97% of their windows and 62% of the other festivals are recognised as outside, at the cost of sending 22% of reference windows away. The Luwian rituals of Taurisa are placed in 12% of

Table 9
Open-set test, windows of 10 words, five folds. Placed: share passing the margin rule (wrongly, for outside texts). AUROC: separation of reference and outside windows by top similarity. Gate: share whose nearest class is an outside genre.

Texts	*n*	placed	AUROC	gate
Cult inventories	300	26%	0.86	89%
Festivals of other traditions	300	60%	0.58	62%
Myths	300	24%	0.79	87%
Historical texts, treaties	300	24%	0.84	97%
Oracles	300	23%	0.88	96%
Prayers and hymns	138	29%	0.81	95%
Instructions	114	33%	0.78	91%
Luwian rituals of Taurisa	14	12%	0.81	95%
Reference traditions (held out)	1,627	68%	–	22%

windows and recognised by the gate in 95%, but rest on only 14 texts.

Applied to the orphans, the gate finds 212 of 2,026 that are closer to an outside genre than to any tradition by more than the margin; 73 of them had been assigned. Most are Hattian-layer placements that look like festivals of other traditions (21) and Kizzuwatna placements that look like myths (16) or instructions (10). They are released as candidates for re-filing.

6. Quantitative Scribal Tradition Mapping

The results suggest a division of labour between two levels of a text, which QTM makes explicit. Borrowing the terms of textual criticism [41, 42], a tradition is treated as an *archetype*: not a reconstructed text, but the body of vocabulary, formulae and foreign-language recitation that its manuscripts share. Each manuscript, and each orphan, is a *witness* that attests a part of it. What identifies the archetype is its content: the layers that carry tradition in Section 5.4 are the cultural ones. What marks the witness is its spelling: plene habits and orthographic rates cluster by composition and copying line, not by tradition, which agrees with the view that Hittite writing habits belong to scribal circles and their schooling [16]. Logograms sit on both sides: the object named is cultural, the decision to write it logographically is scribal [17].

An orphan is a low-coverage witness. Coverage, the share of a tradition's 200 core terms that its assigned orphans attest, is near complete for the traditions that receive many orphans (100% for the Kizzuwatna rituals, 98% for the Hattian layer) and low for those that receive few (28% for the Palaic festivals with 3 orphans). In its present form this measure grows with the number of orphans, so it cannot yet be compared across traditions; rarefaction to equal orphan counts [43], and unseen-species estimates of the kind applied to lost medieval literature [44], are the natural next steps. Read this way, the question becomes historical: which traditions circulated widely in copies that now survive only as fragments, and which are known almost only from their identified compositions.

7. Discussion and limitations

Genre before origin. The residue result shows the main risk of the method. A fragment is placed by what it shares with the reference, and the largest shared layer is genre. A Hittite ritual from outside every reference tradition will look most like the largest ritual class. The open-set test shows the same risk across genres. A placement should therefore be read as "nearest of these" unless the reference includes the plausible alternatives as classes. The genre gate is a first step, and each new tradition narrows the gap: with the Luwian rituals of CTH 758–763 in the reference, 12 residue texts are placed with them rather than with a larger ritual class.

Limitations. TLHdig's normalised forms are uneven across editors, and broken words are dropped, so short fragments over-represent formulaic lines, and their placements rest more on common features (Section 5.5). The idf weights are fitted on the whole reference set. Several traditions are small (Luwian rituals (Puriyanni, Kuwattalla) has 10 test manuscripts), and their recall is uncertain; held out by composition, several of them are not recovered at all (Section 5.2). Festival manuscripts differ by function (outline tablets, day tablets, cult inventories close to festival text [40]), which word profiles may track alongside tradition. The Hattian layer's preference for plene writing and few logograms may reflect Old Hittite manuscripts as much as tradition, and is still to be checked against script dates. The word categories enter only the analysis of what a decision rests on, not the placements; their rules were checked by hand on a sample, but the meaning classes of nouns rest on a gloss list drafted with a language model that is still to be checked. Finally, every placement is a hypothesis for philological verification on the tablet: an orphan placed near a known manuscript is a candidate join, not a join.

8. Conclusion

Content-based assignment places Hittite ritual fragments in literature-defined traditions well above chance, from about ten legible words on, and recovers subgroup structure inside CTH 701 without being a proxy for script date. The signal lies in what is written; spelling marks the copying line. A closed reference over-assigns, and the open-set test shows how to make the abstention honest. QTM turns these observations into a programme: orphans as witnesses, traditions as archetypes, and coverage as the measure of how much of a tradition survives only in fragments.

9. Data and code availability

Code, the frozen CTH 701 label snapshot and all outputs are in the repository https://github.com/shaigordin/measuring-scribal-literacy (folder `orphans/`). `reproduce.sh` downloads TLHdig 0.3 from Zenodo, verifies its checksum and rebuilds every experiment, figure and table; all randomness is seeded and a rerun is byte-identical. The paper itself (`paper/`) is generated from those outputs: `make_numbers.py` writes every number in the text and every results table, and `generated/numbers_index.csv` lists each number with the file and field it comes from.

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Declaration on Generative AI

During the preparation of this work, the author used Claude (Anthropic) in order to: write and run the analysis code, draft the text, and check bibliographic references. After using this tool, the author reviewed and edited the code and the content as needed and takes full responsibility for the publication's content.

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A. Distinctive features per tradition

B. Orthographic profiles

Table 10
The four most distinctive features of each tradition per layer (weighted log-odds with an informative prior [39], on manuscript frequencies), as TLHdig forms. Full lists in `E7_distinctive_features.csv`.

Tradition	Logograms	Hittite vocabulary	Foreign words
Kizzuwatna–Lower Land (Hurrian)	EGIR-ŠÚma, EGIR-ŠUma, UP-NI, BA.BA.ZA	½, manuzia, nupatik, šipantanzi	tai, anaḫi, tea, manuzunna
Syro-Mesopotamia (Hurrian)	AZU, EREN, SÍSKUR, AD.KID	ḫurlili, ḫuprušḫia, aḫrušḫia, aḫrušḫiaz	kelum, kelu, DINGIRna, KI.MIN
Mixed (Hurrian, Syrian, Luwian)	KI.MIN, DINGIRna, NINDA.SIG, EGIRantama	ḫepat, ekuzi, ninatta, tuni	DINGIRna, allai, DUB.SAR, šarra
Luwian festivals (Ištanuwa, Lallupiya)	GAL-ŠU-NU, KA.GAG, ZA.LAM.GAR, DUMU.LUGAL	ḫuḫupal, kimri, anianzi, 9	wašu, anta, paiu, tapala
Ḫuwaššanna festivals	BE-EL, KAŠ, NAR, SÌR-RU	paluaizi, akuanna, takarmun, pianzi	arešti, wa~, ~ziḫpali, ~zuamannia
Palaic festivals	GEŠTINit, UTUi, ŠU.GIma, IMaš	kataḫzipuri, išnuri, ziparua, memaḫḫi	taparnaš, ânta, ni, kuiša
Kizzuwatna rituals	SÍG, ZA.GÌN, ŠU.GI, SA_5	tarpalaš, 7, ḫuittianzi, našma	zuli, enna, ari, 2
Arzawa / western rituals	KAL, KÁ.GAL-TIM, MUŠEN.DÙ, TU_7	kuršaš, paimi, eš, ~ḫi	–
Luwian rituals (Puriyanni, Kuwattalla)	EMEan, SAG.DUi, MUan, ŠU.GIma	pankauaš, karpin, ~ša, ketašša	–
Hattian cult layer	LUGALi, LUGALuš, DUMU.É.GAL, GIŠ.INANNA	ta, ḫalzai, wezi, petai	taparna, pala, katte, ita

Table 11
Mean orthographic rates per tradition as *z*-scores over all reference manuscripts: shares of logograms, Akkadograms, phonetic complements, determinatives, plene spellings, CVC signs and *ú*.

Tradition	logo.	Akkad.	compl.	det.	plene	CVC	*ú*
Kizzuwatna–Lower Land (Hurrian)	+0.24	+0.40	-0.21	+0.04	-0.32	+0.01	-0.30
Syro-Mesopotamia (Hurrian)	+0.08	+0.00	-0.31	-0.12	+0.28	-0.19	-0.31
Mixed (Hurrian, Syrian, Luwian)	+0.26	-0.26	+0.21	+0.29	-0.25	+0.33	-0.10
Luwian festivals (Ištanuwa, Lallupiya)	-0.18	+0.33	-0.12	+0.12	-0.26	+0.03	+0.23
Ḫuwaššanna festivals	+0.36	+0.45	-0.30	+0.35	-0.35	+0.04	-0.16
Palaic festivals	-0.81	-0.20	+0.77	-0.06	+0.15	-0.03	+0.11
Kizzuwatna rituals	+0.13	+0.07	-0.20	-0.46	-0.26	+0.14	+0.10
Arzawa / western rituals	+0.13	+0.26	+0.09	-0.07	-0.34	+0.44	+0.08
Luwian rituals (Puriyanni, Kuwattalla)	-0.12	-0.07	+1.40	-0.56	-0.80	+0.40	+0.12
Hattian cult layer	-0.41	-0.38	+0.27	-0.08	+0.49	-0.16	+0.34